MIDDLEBOROUGH MOBILE PAGING SYSTEM PROJECT

NEW BRUNSWICK COMMUNITY COLLEGE
Business Analysis
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Vendor Evaluation

Attribute / Dimension	Active911	OnPage Pager App	Rapid Reach
Two-way communication & group alerting	Send alerts to individuals, groups, full teams; responders can reply/acknowledge via response buttons.	Secure two-way messaging; supports images/files/voice ; group/team paging + escalation paging.	Responders can confirm/acknowledge ; group and team messaging + chat functionality.
Customizable UI / notifications / workflows / groups / templates	User/device/group management, pre-plans, mapping, hydrants, custom groups, device configuration.	On-call scheduling, escalation rules, alert priorities, alert-until-read, reporting, audit trail, routing logic.	Predefined scenarios, templates, multi-trigger options (app/web/phone/API) , geolocation, fallback and redundancy.
MultiMedia support (images/document s /Voice)	Supports pre-plans, map data, and uploading photos/docs to map markers.	Rich attachments (images, files, voice) with secure encrypted messaging.	Message, chat, and multi-channel alerts supported; website does not explicitly confirm image/document attachments.
Security / encryption / compliance	CAD integration emphasized; detailed encryption/ compliance info not publicly disclosed (requires vendor confirmation).	Encrypted SSL/TLS messaging, audit trails, remote wipe, enterprise-grade security	TLS encrypted communication; supports SaaS or on-prem deployment for data sovereignty.
API / system integration	Compatible with CAD via email protocol or parsing; dispatch → alert → device automation supported	Integrates with enterprise systems (on-call scheduling, alert routing). API details not fully public.	Full Web-service API available; alerts may be triggered via API, alarm panel, phone, web, external triggers.
Deployment speed / redundancy	Cloud-based; integrates with CAD easily; quick setup;	Very fast deployment — system can be	Supports SaaS & on-prem; strong redundancy

Attribute / Dimension	Active911	OnPage Pager App	Rapid Reach
	supports smartphones/desktop s /pagers	activated within minutes; cloud-based with secure audit logging.	(app/SMS/voice/ pager/alarm panel/desktop alerts), notification panel, device monitoring.
Suitability for public safety / first responders	Suitable — widely used by Fire/EMS/Police in small–medium municipalities.	Suitable — used in healthcare, government, on-call teams, IT incident response.	Suitable — high-end pager replacement for emergency response, business continuity, incident management.
Estimated total first-year cost	$18,000 – $28,000	$74,000 – $109,000	$110,000 – $170,000
Estimated annual operating cost (Year 2+)	$5,000 – $8,000/year (lowest)	$50,000 – $72,000/year (medium-high)	$70,000 – $105,000/year (highest)
Overall assessment	Best cost-effectiveness; meets all required functionalities; low risk; ideal for town-size first responders	Strong features but significantly more expensive; suited for hospitals & enterprise on-call teams.	Enterprise-grade, highly redundant, but cost is highest; suited for large or highly regulated agencies.
Recommended choice	✔ Recommended: Active911	Not recommended (cost too high for similar functionality).	Not recommended (enterprise-level cost not justified for town-level needs).

(Note: Functional information 70% from official websites, 15% from google, 15% from ChatGPT; cost figures are estimates for budgeting only.)

Active911 is recommended because it:

- Offers full required functionality (two-way paging, CAD integration, groups, mapping, hydrants, pre-plans).
- Has lowest first-year and long-term cost.
- Is widely used by Fire/EMS/Police across North America.
- Provides faster deployment and lower transition risk from the legacy paging system.
- Matches the town's size and operational model.

Approach

Phase-based Approach

Work Breakdown Structure

1. Project Initiation
   1.1 Kickoff meeting
   1.2 Scope & objective definition
   1.3 Define the stakeholder
   1.4 Project charter creation
   1.5 Project charter approval

2. Project Planning
   2.1 . Schedule & milestones
   2.2 . Budget plan
   2.3 . Risk and communication plan
   2.4 . Resource plan
   2.5 . Control plan
   2.6 . Planning approval

3. Project Execution
   3.1 Requirement gathering
      3.1.1 Requirement workshop
      3.1.2 Technical requirements document
   3.2 . Vendor evaluation & selection
      3.2.1 Request For Information/Quote (RFI/RFQ) preparation
      3.2.2 Vendor evaluation
      3.2.3 Vendor approval
   3.3 Federal application
      3.3.1 Material preparation
      3.3.2 Application approval
   3.4 . System customization & configuration
      3.4.1 Design workshop
      3.4.2 System customization
      3.4.3 System configuration

3.5 . Infrastructure & device setup

3.5.1 Device deployment

3.5.2 User accounts and permissions settings

3.6 . Data migration & testing

3.6.1 Migration preparation

3.6.2 Migration execution

3.6.3 System functional testing

3.6.4 Performance testing

3.6.5 Security testing

3.6.6 User Acceptance Testing (UAT)

3.7 . Training

3.7.1 Create training materials

3.7.2 Rotating staff training

3.8 . Deployment & Go-live

3.8.1 Short-term parallel operation of new and old systems to ensure uninterrupted service

3.8.2 Contingence plan (communication interruption response measures)

3.8.3 System switch

3.8.4 Post go-live monitoring/stabilization

4. Project Closing

4.1 Shutdown old system

4.2 Hand over the project documentation

4.3 Reporting and evaluation

Milestones

Start date: January 02,2026

Finish date: September 18, 2026

Milestones	Completion Date
M1 – Project Initiation Completed	2026-01-15
M2 – Planning completed	2026-01-30
M3 – Requirements Gathering Completed	2026-03-06
M4 – Vendor Selection Finalized	2026-04-16
M5 – Federal Public Safety Network Application Approved	2026-05-19
M6 – Go-Live	2026-08-28
M7 – Project Closure Completed	2026-09-18

Task Estimates

Tasks	Estimated duration (including a 10–20% buffer)
Project Initiation	2 weeks
Project Planning	2 weeks
Requirement Gathering	5 weeks
Vendor Evaluation	3 weeks
Federal Application	4 weeks
Customization & Configuration	8 weeks
Infrastructure & Device Setup	2 weeks
Migration, Testing, UAT	5 weeks
Training	4 weeks 4 weeks (k) First responders work 24/7 on eight-hour shifts, so the training will be conducted in rotating groups)
Deployment & Go-live	2 weeks
Project Closing	1 week

Notes: Operation & Maintenance Continuous (not included in the project cycle)

Sequencing of Tasks

Task	Description	Dependency	Duration
1	Project kickoff and alignment	_	2 days
2	Define scope and object	1	3 days
3	Identify stakeholders & roles	2	2 days
4	Build and approve project charter	3	3 days
5	Develop schedule, milestones	4	3 days
6	Develop detailed budget	5	3 days
7	Risk, communication & resource plan	6	4 days
8	Planning approval	7	1 day
9	Requirements workshop (all departments)	8	4 weeks
10	Technical requirements documents	9	5 days
11	Vendor RFI/RFQ preparation	10	5 days
12	Vendor evaluation	11	18 days
13	Vendor selection approval	12	5 days
14	Federal network approval submission	13	3 days
15	Federal approval	14	4 weeks
16	System design workshop with vendor	13	5 days
17	Application customization	16	4 weeks
18	Application configuration	15,17	4 weeks

Task	Description	Dependency	Duration
19	Server / network preparation	18	5 days
20	Device (mobile app) deployment & provisioning	19	5 days
21	User account & permissions setup	18	3 days
22	Data migration, preparation & mapping	18	3 days
23	Data migration execution	22	3 days
24	Functional testing	20,21	5 days
25	Performance & security testing	24	5 days
26	UAT testing (all responder groups)	23,25	10 days
27	Develop training materials	24	5 days
28	Deliver rotating group training (24/7 staff)	27	4 weeks
29	Parallel operation (old + new)	26,28	7 days
30	System cutover & go-live	29	3 days
31	Post go-live monitoring & stabilization	30	7 days
32	Old system shutdown	31	2 days
33	Final project documentation	32	3 days
34	Final review & evaluation	33	3 days

Resources

Task	Resource	Quantity
Project Initiation/planning	PM, BA, IT leader, Financial staffs	5-7 people
Requirement gathering	BA, stakeholders	6-10 people
Vendor evaluation	IT leader, BA	4 people
System customization	software developer	8-12 people
Network equipment Deployment	Network engineer	2 *7 locations =12 people
Data Migration	Software engineer	5 people
Testing	Software engineer, network engineer, tester	3 Tester 2 *7 locations =14 IT
Training	Training instructors, emergency response	2 instructors (two instructors need to conduct training at 7 locations, so 4 weeks of training is required)

Task	Resource	Quantity
Go-live support	PM, BA, Software engineer, Network engineer, IT leader	2 *6 outside call center locations +4 call center=16 people
Closing	PM, BA	3-4 people

Note: All personnel are from the town government's IT team (350 IT personnel, including 20 personnel directly supporting emergency response).

Assumption Factored into Your Decision for Tasks

- The town council has agreed that overall funding is available, and the approval process will not delay the project timeline.
- The federal public safety network application may take up to four weeks and is expected to be returned within that timeframe.
- The IT department can allocate resources with priority, with no resource conflicts.
- Emergency responders can be trained in rotating groups according to their shift schedule without affecting city operations.
- The network infrastructure is stable enough and does not require major upgrades.
- The data migration volume is moderate, and the old system's log format can be converted automatically.
- The tourist peak season may increase risks, so a 10–15% schedule buffer has been added and distributed across the tasks on the critical path.

Reasonable Task Estimation (including contingency plans)

1. Federal Approval Delays

A 4-week buffer is included in the timeline to account for potential delays in the federal public safety network approval process.

2. System Customization More Complex Than Expected

10%-15% contingency buffer has been reserved on the critical path to cope with additional configuration requirements or unexpected technical challenges during the customization phase.

3. Difficulty Centralizing Emergency Personnel for Training

Training is extended from 3 to 4 weeks, with two instructors training at 7 locations for 4 weeks to support rotating 24-hour shifts without disrupting emergency operations.

4. Critical Defects Discovered During Testing

Additional manpower is allocated during testing, A risk reserve is added, and at least two IT personnel per site will be assigned to provide immediate technical support.

5. Travel Season Impact on Staff Availability

A 10% buffer is applied to UAT and training scheduled from July to August.

IT staffing is increased during UAT, and instructors are deployed on-site to reduce the effect of seasonal staffing shortages.

Summary Tasks

- Project initiation :1.1-1.7
- Project Planning: 2.1-2.3
- Project Execution: 3.1—3.9
- Project Closing: 4.1-4.3

Realistic Resources

Data Source: Official salary data from the Government of Canada (Job Bank)

Data Time: Latest salary statistics in 2024 (still in use in 2025)

Category	CAD	Note
IT Project Manager	$60/hr	Canadian Job Bank median
IT leader	$45/hr	Average
Financial staff	$40/hr	Average
Software Developer	$50/hr	Average
Software Tester	$40/hr	median
Network Administrator / Engineer	$45/hr	average
Business Analyst	$45/hr	median
IT Trainer	$35/hr	average
Fire fighter/Police/Paramedics	$50/hr	Government job salary range
Vendor customization fees	$26000	

Appendix. AON Diagram